

Crossing the Chasm between FinTech and Finance Professionals: A Grounded Analysis of Blockchain, Cloud Computing, and Serverless Development for Future FinTech Information Systems

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Abstract

The purpose of this research is to propose how the future FinTech information systems adopts to emerging technologies such as Blockchain, Cloud Computing, and Serverless Development. This advancement can help finance professionals cross the chasm between FinTech and themselves. For this purpose, we developed a FinTech demo system using those emerging technologies and qualitatively analyze the system using the Grounded Theory and Evidence-Based Practice. Based on the analysis, we propose how to cross the chasm by answering Moore's four fundamental characteristics of FinTech professionals that estrange finance professionals. We argue that the chasm between FinTech and finance professionals can be reduced because the target FinTech system can meet the expectations of finance professions, which are practical, not visionary. The target bank account system is highly secure, scalable, maintainable, and trustworthy, which are the practical references that finance professionals are looking for.

Keywords: FinTech, Blockchain, Cloud Computing, Serverless Development, Grounded Theory, Evidence-Based Practice, Security, Scalability, Maintainability, Trustworthiness, Smart Contract

1. INTRODUCTION

Motivation

Moore mentioned the chasm between the early adopters and the early majority through the revised technology adoption lifecycle in his high-tech marketing model (Moore, 2014, p. 13), which is shown in Figure 1. The early adopters do not rely on well-established references in making their buying decisions, contrary to the early majority. The early adopters are visionaries, but the early majority pragmatists. The early adopters are interested in state of the art technology, but the early majority industry standards. Therefore, the early adopters do not need a support base within a market, but the early majority need a strong support base.

If we compare FinTech and finance professionals, the FinTech professionals are close to the early adopters of emerging technologies in finance while the finance professionals are close to the early majority. Finance professionals are those seeking good references of the emerging technology adoption that are highly supported within their finance market. Moore's revised technology adoption lifecycle in Figure 1 was extended to show the chasm between FinTech and finance professionals. (Check three upward arrows in Figure 1.)

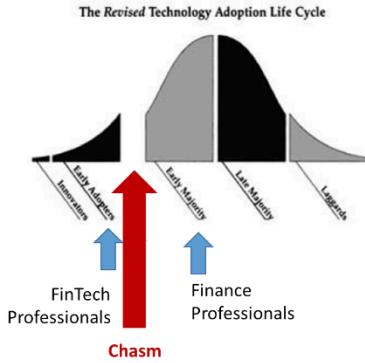


Figure 1. The extended version of the revised technology adoption lifecycle (Moore, 2014, p. 13).

Purpose & Problem Statement

The purpose of this research is to study whether blockchain, cloud computing, and serverless development can reduce the chasm between FinTech and Finance professionals. FinTech professionals are interested in developing new finance information systems using those emerging technologies. However, they are not technologists like Information Technology (IT) professionals. Although finance professionals share some of the ability of the FinTech professionals to use financial-related technologies, they focus on practicality of application of blockchain, cloud computing, and serverless development to finance.

2. BACKGROUND

We start with introducing three key technologies – blockchain, cloud computing, and serverless development. Blockchain is a distributed ledger on peer-to-peer (P2P) network. It helped us develop a Decentralized Application (DApp) on a P2P network, which can harden the integrity of contracts in the DApp and mitigate denial of service (DoS) attacks to the DApp [9]. It also allows us to infuse smart contracts into our software system. Smart contract is a contract that more than two parties agreed and their agreement cannot be changed without mutual agreement. Figure 2 shows a conceptual model of blockchain.

As a solution to this problem, blockchain can be used to execute decentralized applications (DApps). Since DApp works on peer-to-peer networks and data are distributed and maintained in irreversible distributed blockchains, DApp runs as a smart contract with a distributed ledger [2]. However, many existing Apps need to be reengineered to DApps with some guidelines.

Blockchain has several concepts such as distributed ledger, peer-to-peer networking, hash, Merkle tree, public & private key, digital signature, mining, and consensus algorithm. Among those concepts, the recreation professional chooses a key concept such as distributed ledger and designs a group activity for kindergartens. For example, an application of toy supply chain with four role players is designed, which we call ‘Who is the liar?’ game.

1. Iacob, N. M, & Moise, M. L. (2015). Centralized vs. Distributed Database Case Study. *Academic Journal of Economic Studies*, 1(4). p. 119-130.
2. Ahmad, A. (2017). Integration of IoT Devices via a Blockchain-based Decentralized Application. Master Thesis, University of Stuttgart.

Figure 2. A conceptual model of a blockchain

The development of virtualization brought cloud computing to us. Cloud computing is a computational model that supports on-demand computing. Figure 3 shows the list of cloud service providers.

Figure 3. Cloud Service Providers

Serverless development is a cloud computing event-driven model or code execution where infrastructure is abstracted from the developer or end-user (Deloitte, 2018). These applications using serverless computing are run in the cloud. However, these serverless applications that are managed by third-parties are also run on physical servers (Stigler, 2018). Sounds familiar? Cloud is just someone else's computer. So why use serverless computing model then if it's just another physical server? Adopting a cloud-based service can downscale bringing the costs to almost 1/10 compared to maintaining our infrastructure (AWS, 2019d). It provides ease of provisioning, operating, and maintaining. Cloud providers are compliant with most industry standard and regulatory requirements. Cloud helps companies focus more on business logic rather than spend time and resources for provisioning, maintaining, and operating resources (Ayoub, 2012). Amazon Web Services (AWS), Microsoft Azure (Azure), and Google Cloud Platform (GCP).

Serverless applications provide high availability, fault tolerant, scalability, and elasticity. With all the benefits serverless architecture brings, it also has some limitations. These limitations are controlling limit on infrastructure, locked-in with a vendor and the impact of cold start. Moreover, moving the cloud would directly expose our servers to the internet (Ayoub, 2012). In managing our cloud services, the principle of do-not-trust-anyone should be applied. This paper will show how a serverless architecture can impact both security and quality for a web application. The analysis will compare the cloud services of Amazon Web Services (AWS), Microsoft Azure (Azure), and Google Cloud Platform (GCP).

3. RELATED WORK

Grounded Theory Analysis

We apply the Grounded Theory [7, 14] to the artifacts (e.g., source code, diagrams, charts, and logs) of modern web application development projects. The Grounded Theory is an inductive qualitative analysis approach which involves generating a theory (a new discovery) from given source data. The Grounded Theory consists of five steps: first, all data are collected from the test

cases. Second, discrete codes or tags are assigned to the collected data. Third, after assigning discrete codes to the collected data, these codes are then grouped into concepts. Next, the relationships between concepts are examined and classified into categories. Finally, based upon these categories, a new discovery or theory is proposed.

Evidence-Based Practice (EBP)

To select a correct format for non-computing major first-generation-college-bound students, we adopt Evidence-Based Practice (EBP), which came from Medicine. According to Dr. David L. Sackett, who is known as one of the fathers of Evidence-Based Medicine (EBM), EBM is “the conscientious, explicit, and judicious use of current best evidence in making decisions about the care of individual patients. The practice of evidence based medicine means integrating individual clinical expertise with the best available external clinical evidence from systematic research.” (Sackett, et al., 1996; Straus, Glasziou, Richardson, Haynes, & Sackett, 2011). Software Engineering research community adopted EBM and proposed Evidence-Based Software Engineering (EBSE) (Kitchenham, Dyba, & Jorgensen, 2004; Dyba, Kitchenham, & Jorgensen, 2005).

4. APPROACH

To answer how blockchain and cloud technology can reduce the chasm between FinTech and Finance professionals, we demonstrate and analyze whether blockchain and cloud technology can make a finance information system that is more practical in terms of security, scalability, maintainability, and smartness. First, we develop a simple new finance information system, called Savings Bank Account Management, using blockchain, cloud, and serverless computing. Second, we modeled the architectures of the new finance information system to collect data that both FinTech and finance professionals can understand in a high-level abstraction. Third, we apply the Grounded Theory to the targets system: 1) Discrete codes for technologies employed are assigned to the collected data. 2) These codes are grouped into concepts. 3) The relationships between concepts are examined and classified into categories. 4) Finally based upon these categories, a new discovery is proposed. Lastly, we use the architectural model and the outcome of the grounded theory to discuss how the chasm between visionary FinTech and practical finance professionals can be reduced. This is with the help of the target system using blockchain, cloud, and serverless computing that meets the expectations of finance professions.

5. A USE CASE FOR A FINTECH DEMO SYSTEM

We first propose a use case for a Savings Bank Account Management. For example, Jon Doe opened his saving account at ABC bank with \$1,000 on December 10, 2018. The annual interest rate was 2% per year. He withdrew \$500 on January 15, 2019. Then, he deposited \$200 in February 4, 2019. He withdrew \$600 on March 15, 2019. Then, he had a deposit with \$ 900 on April 23, 2019. Jon checked his balance on April 30, 2019. Figure 2 shows the use case view for the demo FinTech system.

Figure 2. A Use Case View of the Demo FinTech System

Based on the given use case, ABC bank can have the following transactions for Mr. Jon Dow, which is shown in Table 1. It shows eight data points captured including transaction id, date, type, previous balance, the amount to be handled, the number of the elapsed days between the previous and the current transaction, the interested earned during the elapsed days, and the current balance.

Table 1. Example Transactions of a Savings Account at a Bank ABC

ID	Date	Type	Previous Balance	Amount	Days	Interest	Current Balance
1001	12/10/2018	OPENING	\$0.00	\$1,000.00		\$ -	\$1,000.00
1002	1/15/2019	WITHDRAW	\$1,000.00	(\$500.00)	35	\$ 1.92	\$501.92
1003	2/4/2019	DEPOSIT	\$501.92	\$200.00	19	\$ 0.52	\$702.44
1004	3/15/2019	WITHDRAW	\$702.44	(\$600.00)	38	\$ 1.46	\$103.90
1005	4/23/2019	DEPOSIT	\$103.90	\$900.00	38	\$ 0.22	\$1,004.12
1006	4/30/2019	CHECK	\$1,004.12	\$0.00	6	\$ 0.33	\$1,004.45

6. A DEMO FITECH SYSTEM

We develop a demo FinTech system called Savings Bank Account Management System (SBAMS) by using three important technologies in finance – Blockchain, Cloud, and Serverless Computing. First, we start with a legacy SBAMS without using any FinTech. Then, we modernize this legacy system to the demo FinTech system using those technologies.

The legacy system uses web technology. Although the target system uses the same technology, the legacy employs a centralized application and database servers for clients based on client/server (C/S) computing. Figure 3 shows the deployment view of the legacy system. The legacy system uses the current web technology using client/server (C/S) computing. A browser sends its requests to a dedicated server of the ABC bank and the server responds according to the requests. When a bank customer uses an online banking facility of the bank, the client accesses his or her account by providing login credentials that are stored in a database server through its application server. The bank application server also acts as a client and requests user details from

the bank database server. If the credentials are correct, the bank application server responds the requests according to the business logic implemented and the bank customer's browser receives relevant results in his or her web browser.

Figure 3. The Deployment View of the Legacy System

The legacy system in Figure 3 has built-in weaknesses. First, this centralized nature of C/S computing brings a fatal problem to the application – security. DDoS attack can stop their business to customers (availability). Insider attackers can change the centralized transaction information (integrity). Hackers can illegally access data due to authentication systems using weak user ID and password (confidentiality). Second, the bank has limit in terms of scaling up or down their transaction services for their peak or idle time if the bank operators own those C/S systems without using a cloud infrastructure. The IT department of the bank has to maintain all applications on their systems as components, not as microservices. Moreover, the Bank can change any contract with students like interest rate or service fee without getting any consensus from their partners.

We modernize this legacy system into the target demo FinTech system by adopting blockchain, cloud, and serverless computing. The target system in Figure 4 uses web technology that the legacy used. However, this target system uses both C/S and peer-to-peer (P2P) computing. The private blockchain is built on the P2P computing paradigm. Also, contracts between the bank and its partners are stored into the blockchain. Also, the target system is deployed onto the bank's cloud system. The target system is developed by using serverless development. Many services are borrowed from the service providers.

Figure 4. The Deployment View of the Target FinTech System

7. QUALITATIVE ANALYSIS WITH THE GROUNDED THEORY

7.1 Data Collection

We collect data from the target demo FinTech system - SBAMS. For implementation, we use Amazon blockchain, cloud computing, and serverless development. The system consists of 10 components: Client Web Interface, Identity Access Management, Backend API Services & Management, Static Content Hosting, Bank Employee Account Management, Bank Transaction DB, Savings Account, Bank Customer Account Management, Cloud Management Application, and Monitoring and Management Service. Table 2 shows the collected data.

7.2 Discrete Codes/Tags

We assign discrete codes or tags to the collected data of the target demo FinTech system. For a discrete code, we check which tools, engines, or libraries are used to demonstrate the ability of creating the data. Table 2 shows the relationship between data and the code. For example,

7.3 Classified Concepts

We group the discrete codes into concepts in Table I2, in which a concept is based upon a skill that is required to demonstrate abilities of using tools, engines, and libraries. For example, .

7.4 Concept Categorization using EBP

We construct categories that highlight the relationship between concepts. We use properties that finance professionals are looking for such as security, scalability, trustworthiness, smartness, and maintainability. To identify properties, we use EBP. For example, The KSAs and DevOps are summarized in Table 1.

Severless –no provision, no operation,
Cloud (EC2) – provision, operation

Table 2. FinTech and Finance Professionals

Views from FinTech Professionals		Views from Finance Professionals	
Data (components)	Discrete Code (technologies)	Concepts (skills)	Categorization (properties)
Client Web Interface	JavaScript, ReactJS 16.8.3, Material UI 3.2	Single-Page Application	Maintainability
Identity Access Management	AWS IAM	Cloud - Serverless	Security - C
Backend API Services & Management	AWS EC2 with Node.js 11.6, Express 4.16.4, Helmet 3.17.0	Cloud - IaaS	- Scalability - Maintainability
Static Content Hosting	AWS S3	Cloud - Serverless	Scalability

TEMPLATE

			• Maintainability • Security - A
Bank Employee Account Management	• AWS Cognito User Pool	• Cloud - Serverless	• Scalability • Maintainability – low operation cost • Security - CIA
Bank Transaction DB	• AWS Managed Blockchain based on Hyperledger Fabric 1.2	• Private Blockchain	• Security – CIA • Maintainability
Savings Account	• Fabric Client Node with Node.js 11.6 • AWS EC2	• Private Blockchain	• Trustworthiness - Smart Contract
Cloud Management Application	• AWS Management Console	• Cloud - PaaS	• Maintainability
Monitoring and Management Service	• AWS CloudWatch	• Cloud - SaaS	• Maintainability

8. FINDINGS AND DISCUSSIONS

Finally, we propose a new discovery from Table2 above: A target FinTech system using Blockchain, Cloud Computing, and Serverless Computing *support the perspectives of finance professionals who want to be early majorities*. Through the four steps of the Grounded Theory analysis, we discover what abilities are required in creating the artifacts of the modern web application development projects, which skills are exercised to demonstrate the abilities, what knowledge is studied to understand the concepts of the skills, and then how the knowledge is related to the four properties of the views of finance professionals.

We also answer four fundamental characteristics of visionaries i.e FinTech professionals that alienate pragmatists i.e. finance professionals, which Moore mentioned in his book, based on our findings (Moore, 2014, p. 41-43).

1. 3

9. CONCLUSION

Key technologies in FinTech such as Blockchain and Cloud can help FinTech and finance professionals cross the chasm between themselves. The state of the art technology, which the FinTech professionals advocate, can encourage finance professionals to be the early majority of adopting blockchain and cloud technology with better understanding: 1) The architectural model, the higher abstraction of the target system, can help both professionals to have better understanding of both sides (comprehensibility). 2) The distributed ledger on the P2P

infrastructure hardens Distributed Denial of Service (DDoS) attacks (security - availability) and any data changes are traced (security - integrity) with encrypted transaction (security - confidentiality). 3) The blockchain in Amazon cloud can be scaled up and down based on demand (scalability). 4) The serverless development of web app can reduce the cost of server maintenance and focus on business domain (maintainability). 5) The smart contract between stakeholders can be implemented in code and executed on the blockchains (smart), which is not available in the current finance information system not using blockchain.

10. FUTURE WORK

[Describe what needs to be done more in the near future if you have more time.]

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